

Web Technology I

Unit 3 - HTML 5 and Features

Unit 3. HTML 5 and Features [5 Hrs.]

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3.1. Introduction

<https://www.w3docs.com/learn-html/html5-introduction.html>

https://www.tutorialspoint.com/html5/html5_overview.htm

HTML 5 is the fifth and current version of HTML. It has improved the markup available for documents and has introduced application programming interfaces (API) and Document Object Model (DOM).

HTML5 is not only a new version of HTML language enriched with new elements and attributes, but a set of technologies for building more powerful and diverse web sites and applications, that support multimedia, interact with software interfaces, etc.

Browser Support - The latest versions of Apple Safari, Google Chrome, Mozilla Firefox, and Opera all support many HTML5 features and Internet Explorer 9.0 will also have support for some HTML5 functionality. The mobile web browsers that come pre-installed on iPhones, iPads, and Android phones all have excellent support for HTML5.

[

HTML5 is an advanced version of HTML. It contains all the features of HTML along with new tags like <video> , <audio> and <canvas>. There are also tags for

including semantics, and these tags are <article>, <header>, <footer> etc. HTML5 improves the HTML to support the latest multimedia and new features, including graphics, music, etc.

]

Features:

HTML5 introduces a number of new elements and attributes that can help you in building modern websites. Here is a set of some of the most prominent features introduced in HTML5.

- **New Semantic Elements** - Enrich semantic content by including <header> <footer>, <article>, <section> and <figure> are added.
- **Audio & Video** - It has introduced new multimedia features which supports both audio and video controls by using <audio> and <video> tags. You can embed audio or video on your webpages without resorting to third-party plugins.
- There are new graphics elements including vector graphics and tags.
(**Canvas** - This supports a two-dimensional drawing surface that you can program with JavaScript.)
- Allows drawing various shapes like triangle, rectangle, circle, etc.
- **Drag and Drop**- The user can grab an object and drag it further dropping it to a new location. Drag and drop the items from one location to another location on the same webpage.
- **Geo-location services**- It helps to locate the geographical location of a client. Visitors can choose to share their physical location with your web application.
- Web storage facility which provides web application methods to store data on the web browser. . (**Persistent Local Storage** - To achieve without resorting to third-party plugins.)
- Uses SQL database to store data offline.
- Capable of handling incorrect syntax.
- Easy DOCTYPE declaration i.e., <!doctype html>
- Easy character encoding i.e., <meta charset="UTF-8">
- **WebSocket** - A next-generation bidirectional communication technology for web applications.
- **Server-Sent Events** - HTML5 introduces events which flow from web server to the web browsers and they are called Server-Sent Events (SSE).

Advantages:

- All browsers supported.
- More device friendly.
- Easy to use and implement.
- HTML 5 in integration with CSS, JavaScript, etc. can help build beautiful websites.

Disadvantages:

- Long codes have to be written which is time consuming.
- Only modern browsers support it.

Note –

DOM

<https://www.geeksforgeeks.org/dom-document-object-model/>

Document Object Model (DOM) is a programming interface for HTML(HyperText Markup Language) and XML(Extensible markup language) documents. It defines the logical structure of documents and the way a document is accessed and manipulated.

DOM is a way to represent the webpage in a structured hierarchical way so that it will become easier for programmers and users to glide through the document. With DOM, we can easily access and manipulate tags, IDs, classes, Attributes, or Elements of HTML using commands or methods provided by the Document object. Using DOM, the JavaScript gets access to HTML as well as CSS of the web page and can also add behavior to the HTML elements. so basically Document Object Model is an API that represents and interacts with HTML or XML documents.

Example –

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <link rel="stylesheet" type="text/css" href="style.css">
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
    <title></title>
  </head>
  <body>
    <article>HTML5</article>
    <section>It is advanced HTML<section>
    <section>Supports multimedia features<section>
  </body>
</html>
```

3.2. Difference between HTML and HTML 5

<https://www.javatpoint.com/html-vs-html5>

<https://www.simplilearn.com/tutorials/html-tutorial/html-vs-html5>

<https://www.scaler.com/topics/difference-between-html-and-html5/>

Assignment



Difference Between Html and Html5

Html	Html5
Doctype declaration in Html is too longer <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">	DOCTYPE declaration in Html5 is very simple <!DOCTYPE html>
character encoding in Html is also longer <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">	character encoding (charset) declaration is also very simple <meta charset="UTF-8">
Audio and Video are not part of HTML4	Audio and Videos are integral part of HTML5 e.g. <audio> and <video> tags.
Vector Graphics is possible with the help of technologies such as VML, Silverlight, Flash etc	Vector graphics is integral part of HTML5 e.g. SVG and canvas
It is almost impossible to get true GeoLocation of user browsing any website especially if it comes to mobile devices.	JS GeoLocation API in HTML5 helps identify location of user browsing any website (provided user allows it)
Html5 use cookies.	It provides local storage in place of cookies.
Not possible to draw shapes like circle, rectangle, triangle.	Using Html5 you can draw shapes like circle, rectangle, triangle.
Does not allow JavaScript to run in browser. JS runs in same thread as browser interface.	Allows JavaScript to run in background. This is possible due to JS Web worker API in HTML5
Works with all old browsers	Supported by all new browser.

3.3. HTML 5 New Semantics Elements (HEADER, FOOTER, SECTION)

<https://www.w3docs.com/learn-html/semantic-elements-in-html5.html>

Semantics is the study of the meanings of words and phrases in language. Semantic elements are elements with a meaning.

- A semantic element clearly describes its meaning to both the browser and the developer.
- Examples of **non-semantic** elements: <div> and - Tells nothing about its content.
- Examples of **semantic** elements: <form>, <table>, and - Clearly defines its content.

Semantic elements are one of the most significant introductions in HTML5. In the previous versions of HTML, the generic <div> tag with an id or class attribute was used for structuring a web page. For example, for defining sidebars, footers, menu or other structural blocks, the <div> tag was used with the corresponding meaning (div class="footer").

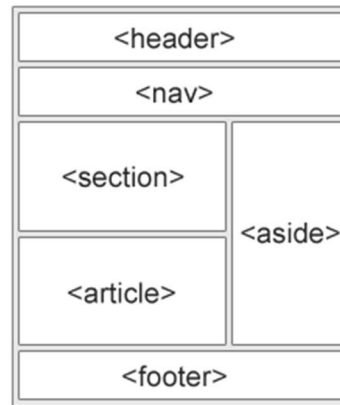
Semantic elements in HTML have intrinsic meaning and convey that meaning both to the browser and the developer. They clearly define what kind of content they contain (for example, the <footer> tag is used instead of <div id="footer">).

New Semantic Elements in HTML5

Many web sites contain HTML code like: `<div id="nav">` `<div class="header">` `<div id="footer">` to indicate navigation, header, and footer.

HTML5 offers new semantic elements to define different parts of a web page:

- `<article>`
- `<aside>`
- `<details>`
- `<figcaption>`
- `<figure>`
- `<footer>`
- `<header>`
- `<main>`
- `<mark>`
- `<nav>`
- `<section>`
- `<summary>`
- `<time>`



`<header>` element

<https://www.w3docs.com/learn-html/html-header-tag.html>

The `<header>` element defines a header for the document or section. It usually contains a logo, search bar, navigation links, etc.

This tag doesn't present a new section in the outline. A `<header>` element commonly contains the heading (an `<h1>`–`<h6>` element) of a surrounding section. However, this is not required.

The `<header>` tag is one of the HTML5 elements. In an HTML document, it is allowed to use several `<header>` tags, which can be placed in any part of it.

Note –

It is not permitted to place the `<header>` tag inside the `<footer>` and `<address>` elements, and in another `<header>` tag.

Syntax –

`<header>` Write your content here.`</header>`

Example –

Page Header

The example code snippet below shows the use of header tags to display the heading of an HTML document:

Example:

```
<header>
  
  <h1>Page Title</h1>
</header>
```

Article Header

The example code snippet below shows the use of header tags to display the heading of an article:

Example:

```
<article>
  <header>
    <h2>An Example of the HTML Header Tag </h2>
    <p>Posted on Sunday, <time datetime="2012-05-01">1 May
2012</time> by Alex</p>
  </header>
  <p>This is an sample text placed within the document.</p>
  <p><a href="#">Read more....</a></p>
</article>
```

<footer> element

<https://www.w3schools.in/html5/footer-tag>

The <footer> element defines the footer of a web page or a section. As a rule, it contains copyright information, contact details, navigation links, etc.

You can have several <footer> tags on a web page. For example, you can place a footer inside the <article> tag to store information related to the article (links, footnotes, etc.).

Note -

The tag can contain other HTML elements, except for the <footer> and <header> tags.

If the footer contains contact details, you should put it in the <address> tag.

Syntax –

<footer> Write your content here. </footer>

Example –

```
<footer>
  <p>© 2009 — 2022 W3schools® of Technology.</p>
</footer>
```

Extra -

Char	Number	Entity	Description
©	©	©	COPYRIGHT SIGN
®	®	®	REGISTERED SIGN
€	€	€	EURO SIGN

Example –

```
<!DOCTYPE html>
<html>
<body>

<h1>HTML Entity Example</h1>

<footer>The copyright symbol: &copy;</footer>

</body>
</html>
```

Output –

HTML Entity Example

The copyright symbol: ©

<section> element

<https://www.w3docs.com/learn-html/html-section-tag.html>

The <section> element is used to group standalone sections within a webpage containing logically connected content (news block, contact information etc.). The <section> tag is often used when creating a landing page to divide the page into separate logical blocks.

Note - Don't use the <section> tag as a universal container for creating page structure; you should use the <div> tag for that purpose.

Syntax –

<section> Write your content here. </section>

Example –

```
<!DOCTYPE html>
<html>
  <head>
    <title>Using the section tag</title>
  </head>
  <body>
    <section>
      <h1>HTML</h1>
      <p>HTML-Hypertext markup language is the standard markup language for
creating web pages and web applications. </p>
    </section>
    <section>
      <h1>CSS</h1>
      <p>Cascading Style Sheets </p>
    </section>
  </body>
</html>
```

Output –

HTML

HTML-Hypertext markup language is the standard markup language for creating web pages and web applications.

CSS

Cascading Style Sheets

Extra Note –

Some new tags have been introduced with HTML5 for better structure of HTML documents:

Tag	Description
-----	-------------

nav	The navigation tag denotes a document section in an HTML document designed to contain navigation links.
header	The header tag defines the header of the HTML document or section. It usually contains a logo, navigation, search, etc.
section	The section tag defines a section in an HTML document.
article	The <article> tag is one of the HTML5 element used to specifies independent, self-contained content in an HTML document.
aside	The aside tag is one of the HTML5 elements that define a section containing additional information related to the content surrounding the "aside" element.
dialog	The dialog tag is used to specify a conversation section such as popup dialogs and modals in an HTML document.
figure	The figure tag is used to display a caption of embedded images or videos.
footer	The footer tag defines the footer of the HTML document or section. It usually contains document copyright, navigation, search, etc.

3.4. HTML 5 New Elements

3.4.1. Tables, Images, Colors, Canvas, Forms

3.4.2. Interactive Elements

3.4.3. Graphics

3.4.4. Multimedia

The most interesting new elements are:

New **semantic** elements like <header>, <footer>, <article>, and <section>.

New form **control attributes** like number, date, time, calendar, and range.

New **graphic** elements: <svg> and <canvas>.

New **multimedia** elements: <audio> and <video>.

3.4.1. Tables, Images, Colors, Canvas, Forms

Tables

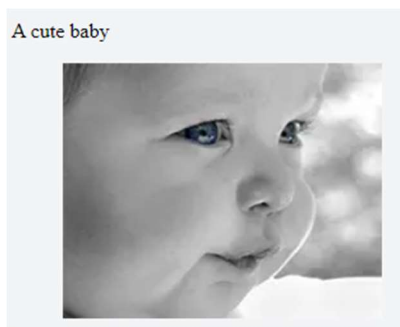
-- same as html

Images

<figure> tag - Defines self-contained content, like illustrations, diagrams, photos, code listings, etc.

Example -

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title of the document</title>
  </head>
  <body>
    <p>A cute baby</p>
    <figure>
      
    </figure>
  </body>
</html>
Output –
```



<figcaption> - Defines a caption for a <figure> element.

Example -

```
<!DOCTYPE html>
<html>
  <head>
    <title>Title of the document</title>
  </head>
  <body>
    <p>Image of a baby</p>
    <figure>
```

```

<figcaption>Fig.1 – Cute baby</figcaption>
</figure>
</body>
</html>
```

Output –



Colors

Canvas

<https://www.javatpoint.com/html-canvas>

The **HTML canvas element** provides HTML a bitmapped surface to work with. It is used to draw graphics on the web page.

The **HTML 5 <canvas> tag** is used to draw graphics using scripting language like JavaScript.

The <canvas> element is only a container for graphics, you must need a scripting language to draw the graphics. The <canvas> element allows for dynamic and scriptable rendering of 2D shapes and bitmap images.

A canvas is a rectangle like area on an HTML page. It is specified with canvas element. By default, the <canvas> element has no border and no content, it is like a container.

```
<canvas id = "mycanvas" width = "200" height = "100"> </canvas>
```

HTML 5 Canvas Tag Example

```
<canvas id="myCanvas1" width="300" height="100" style="border:2px solid;">
```

Your browser does not support the HTML5 canvas tag.

```
</canvas>
```

Example –



Example –

```
<canvas id="myCanvas" width="250" height="150" style="border:1px solid #c3c3c3;">
```

Your browser does not support the HTML5 canvas tag.

```
</canvas>
```

```
<script>
```

```
var c = document.getElementById("myCanvas");
```

```
var ctx = c.getContext("2d");
```

```
ctx.fillStyle = "#FF0000";
```

```
ctx.fillRect(0,0,200,100);
```

```
</script>
```

Output:



Forms -

<https://www.w3schools.in/html/web-forms>

Common types of web forms are - Contact form, Registration form, Sign-in Form, Order form, Survey form, Search form, etc.

The <form> tag in HTML is used to define the form in an HTML document. This tag is a container for all other input elements used within this form tag to obtain user input.

Syntax:

```
<form>
<!-- HTML Form Element -->
</form>
```

Attributes of the HTML <form> Element

Syntax:

```
<form action="signin.php" method="post">
<!-- HTML Form Element -->
</form>
```

Example -

```
<form name="SignIn" method="post" action="signin.php" >

  <label for="Email">Email:</label><br>
  <p><input type="email" name="email" id="Email"></p>

  <label for="Password">Password:</label>
  <p><input type="password" name="password" id="Password"></p>

  <input type="submit" value="Sign in">

</form>
```

Output –

The image shows a web form with the following elements:

- Email:** A text input field.
- Password:** A password input field.
- Sign in**: A submit button.

Annotations in the image:

- An arrow labeled "HTML Form" points to the entire form container.
- An arrow labeled "HTML Form Elements" points to the input fields and the submit button.

HTML5 introduced more types of input elements. They are:

Input Type	Description	HTML Markup
number	It defines a single-line text field for only numbers.	<input type="number">
tel	It defines a single-line text field for telephone numbers.	<input type="tel">
range	It displays a slider to set the value of an input element.	<input type="range">
url	It defines a single-line text field for a URL address. The input value is automatically verified to ensure a correctly formatted URL.	<input type="url">
search	It defines a single-line text field that is styled differently by web browsers.	<input type="search">
color	It provides an interface to select colors using a visual color picker.	<input type="color">
date	It displays a calendar to select a date that is styled differently by different web browsers.	<input type="date">
time	It specifies single-line text fields for the time input (hours, minutes, and seconds optionally) format.	<input type="time">
datetime-local	It displays a calendar to select both date and time.	<input type="datetime-local">
month	It displays a calendar to select a month.	<input type="month">
week	It displays a calendar to select a week.	<input type="week">

3.4.2. Interactive Elements

Some elements have built-in interactivity, meaning they do something when you interact with them. A common example of this is links, which respond when you hover or click, but there are some new elements that offer other kinds of interactivity.

Users are able to interact with interactive tags such as AUDIO (which allows an audio file to be played on an HTML page), VIDEO, and IMG because these tags were created so that users could interact with them.

HTML Interactive Tags List

The various interactive tags available in HTML are outlined in the table that can be found below.

Tags	Description
A	represents a link in an HTML document.
AUDIO (if the controls attribute is present)	represents audio and sound streams.
BUTTON	represents a push button.
DETAILS	provides additional information or control for users.
EMBED	represents the plugin content in an HTML document.
IFRAME	represents an inline frame.
IMG (if the usemap attribute is present)	represents an image.
INPUT (if the type attribute is not in the hidden state)	represents an input control.
KEYGEN	represents a control to generate a key pair.
LABEL	represents the label for an input tag.
MENU (if the type attribute is in the toolbar state)	represents a menu list.
OBJECT (if the usemap attribute is present)	represents an embedded object.
SELECT	represents a select list (drop-down list).
TEXTAREA	represents a multi-line text input control.
VIDEO (if the controls attribute is present)	represents videos or movie streams.

HTML Interactive Tags Example

Now is the time to give an example of some of the interactive tags listed in the above table. For example, the "A" tag comes under the category of interactive tags because

the user can interact with the content of this tag as shown in the example given below:

HTML Code

```
<!DOCTYPE html>
<html>
<body>

    <p>Visit <A href="https://codescracker.com">codescracker.com</A></p>

</body>
</html>
```

Output

Visit codescracker.com

If the user clicks on the text "codescracker.com," then he or she will be redirected to the URL "https://codescracker.com" on the web.

The following example uses the "BUTTON" tag to create a button on the web so that we can perform a particular action when the button is clicked.

HTML Code

```
<!DOCTYPE html>
<html>
    <head>
        <script>
            function codescracker() {
                alert("Hey, what's Up?")
            }
        </script>
    </head>
<body>

    <button onclick="codescracker()">Click Me</button>

</body>
</html>
```

Output



Details & Summary tag –

```
<details>
    <summary>How much wood could a woodchuck chuck?</summary>

    <p>A woodchuck would chuck as much wood as a woodchuck
could chuck</p>

</details>
```


o/p

► How much wood could a woodchuck chuck?

When user click here than -

▼ How much wood could a woodchuck chuck?

A woodchuck would chuck as much wood as a woodchuck could chuck

3.4.3. Graphics

Elements	Description
<code><canvas></code>	Defines an area on the web page, where we can create different objects, images, animations, photo compositions via scripts.
<code><svg></code>	Draws scalable vector graphics.

http://www-db.deis.unibo.it/courses/TW/DOCS/w3schools/html/html5_svg.asp.html#gsc.tab=0

SVG

SVG stands for Scalable Vector Graphics. SVG is used to define graphics for the Web.

<svg> Element

The HTML `<svg>` element (introduced in HTML5) is a container for SVG graphics.

SVG has several methods for drawing paths, boxes, circles, text, and graphic images.

Example –

```
<!DOCTYPE html>
<html>
<body>

<svg width="100" height="100">
  <circle cx="50" cy="50" r="40"
    stroke="green" stroke-width="4" fill="yellow" />
  Sorry, your browser does not support inline SVG.
</svg>

</body>
</html>
```

Output –



Assignment –

Differences Between SVG and Canvas

- SVG is a language for describing 2D graphics in XML.
- Canvas draws 2D graphics, on the fly (with a JavaScript).
- SVG is XML based, which means that every element is available within the SVG DOM. You can attach JavaScript event handlers for an element.
- In SVG, each drawn shape is remembered as an object. If attributes of an SVG object are changed, the browser can automatically re-render the shape.
- Canvas is rendered pixel by pixel. In canvas, once the graphic is drawn, it is forgotten by the browser. If its position should be changed, the entire scene needs to be redrawn, including any objects that might have been covered by the graphic.

3.4.4. Multimedia

<https://www.w3docs.com/learn-html/audio-and-video-in-html5.html>

https://www.w3schools.com/html/html_media.asp

Elements	Description
<code><audio></code>	Indicates the variations of the same audio file.
<code><embed></code>	Is used as a container for external applications, multimedia and interactive content that the browser doesn't understand.
<code><source></code>	Defines multiple media resources in different format: video, audio or image.
<code><track></code>	Specifies text tracks for media elements.
<code><video></code>	Embeds video on an HTML document.

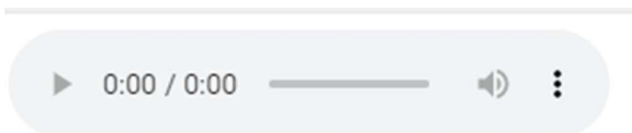
`<audio>` element

To play an audio file in HTML, use the **`<audio>`** element:

Example

```
<audio controls>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

Output-



Attributes

= New in HTML5.

Attribute	Value	Description
-----------	-------	-------------

autoplay	autoplay	Specifies that the audio will start playing as soon as it is ready
--------------------------	----------	--

controls	controls	Specifies that audio controls should be displayed (such as a play/pause button etc)
--------------------------	----------	---

loop	loop	Specifies that the audio will start over again, every time it is finished
----------------------	------	---

muted	muted	Specifies that the audio output should be muted
-----------------------	-------	---

preload	auto metadata none	Specifies if and how the author thinks the audio should be loaded when the page loads
-------------------------	--------------------------	---

src	URL	Specifies the URL of the audio file
---------------------	-----	-------------------------------------

Example –

```
<audio controls autoplay>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
```

```
<audio controls muted>
  <source src="horse.ogg" type="audio/ogg">
  <source src="horse.mp3" type="audio/mpeg">
  Your browser does not support the audio tag.
</audio>
```

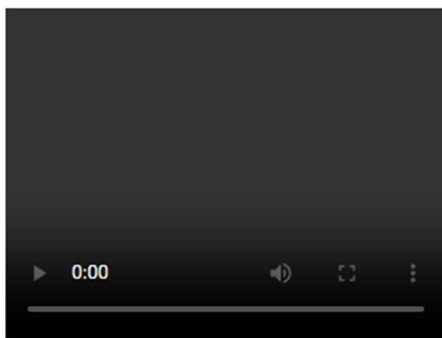
<video> Element

To show a video in HTML, use the **<video>** element:

Example

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

Output –



To start a video automatically use the **autoplay** attribute:

Example

```
<video width="320" height="240" autoplay>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

Extra Note –

Adding New Elements to HTML

You can also add any new element to HTML with a browser trick.

This example adds a new element called **<myHero>** to HTML, and defines a display style for it:

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Creating an HTML Element</title>
  <script>document.createElement("myHero")</script>
  <style>
    myHero {
      display: block;
      background-color: #ddd;
      padding: 50px;
      font-size: 30px;
    }
  </style>
</head>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>

<myHero>My First Hero</myHero>

</body>
</html>
```

Output -

My First Heading

My first paragraph.

My First Hero

The JavaScript statement **document.createElement("myHero")** is added, only to satisfy IE.